

P5 Champion Projects — Content & User Flows

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Context

In P5 (age 10), each child raises one AI champion — a humanoid robot — by completing 17 short projects, one per lesson. Each project gives the champion one durable ability tied to that lesson's AI concept. The child never types code: they configure and tune a pre-built module by supplying the real content that matters (their own photos, rules, thresholds, judgment calls). A named AI buddy coaches them and, on the child's approval, helps tune.

One rule shapes every project: the pre-built part is only the invisible engine; the child supplies all the real content, from the real world; the buddy guides them to do the real job but never supplies the content. This doc gives each project's content and the child's step-by-step flow.

How to read each project

Every project runs the same six beats — **Hunt** (gather real content) → **Teach** (feed it in) → **Come alive** (a champion part lights up) → **Prove** (test on new input) → **Show off** (the child's ability runs a real job on our pre-made City stage) → **Keep** (the ability stays). Only the specifics are listed:

The AI City is a stage we build (districts, the items to sort, the streets, the buses). The child supplies only the *ability* that performs in it. "Show off" = their trained ability doing a real job on our stage.

- **Concept** — what the lesson taught.
- **Endowment** — the ability the champion gains.
- **Engine** — the pre-built, invisible part.
- **Real-life job** — the content the child supplies.
- **Tuning surface** — the decisions the child makes.
- **User flow** — the six beats, made concrete.
- **Done when** — the success bar, always on fresh (unseen) input.
- **Buddy moments** — two sample lines.
- **City district** — where it demos.
- **Watch-outs** — safety/resilience notes.

Global defaults (all 17): deterministic sim; no personal data (nicknames/first names only, camera images kept in memory); flash-safe; tap always works; the champion part lights up at Come alive; the ability saves at Keep.

Content decisions to confirm: for the sim-heavy projects (L2, L11, L14, L13) the child authors the test scenario — they build the thing the champion must handle. For L7/L9 they run a light class/home survey (times only, anonymized). For L16 they gather real feedback. These keep every project child-owned, not just the camera ones.

Cluster A — SENSES

L1 · Recycle-Eye — *name an object's material*

- **Concept:** an image classifier learns from labeled examples.
- **Endowment:** the champion can look at any object and name its recyclable material.
- **Engine:** image classifier (MobileNet+KNN); annotate-and-train UI; held-out accuracy + confusion display; a test bench on unseen items. **No photo bank.**
- **Real-life job:** a **recycling scavenger hunt** — photograph 5+ real items around the room/home (their bottle, their crisp packet) and sort them into class buckets.
- **Tuning surface:** which/how many examples per class; resolve confusions (foil ↔ can); invent a new class; set the "unsure → human bin" threshold.
- **User flow:** **HUNT** snap 5+ real items → **TEACH** train; accuracy climbs live → **COME ALIVE** the eye lens powers on, champion names a held object → **PROVE** test bench shows a brand-new object → **SHOW OFF** sort the City recycling belt; sharpest-eye + coolest-invented-class rivalry → **KEEP** lens on the head.
- **Done when:** ≥90% on the held-out set **and** ambiguous items go to the human bin.
- **Buddy moments:** *"Foil and cans mix up, and you've got 2 foil vs. 9 can — go snap 4 more foils at odd angles?" · "91%! But it confidently guessed on that squished cup — add an 'unsure' bin so it asks a human?"*
- **City district:** Smart Recycling Sorter.
- **Watch-outs:** camera → runs on `http://localhost`; no camera = honest block + retry.

L2 · Deep-Scan Sense — *identify hidden hazards from noisy signals*

- **Concept:** tune a scanner's speed↔accuracy trade-off; train a signal-shape classifier.
- **Endowment:** the champion can scan hidden ground and flag hazards from noisy charts, fast enough to beat the clock.
- **Engine:** scanner grid sim (seeded noisy signals); live Speed-vs-Accuracy graph; chart→hazard classifier; a timer.
- **Real-life job:** the child **buries their own hazard field** — places pipes/rocks/cavities on a plot — then must find them; and **labels a few chart signatures** (spiky = pipe, broad = cavity) to teach the Sense.
- **Tuning surface:** sweep-speed & sensitivity sliders to hit *fast-enough* + *>90% accuracy*; the chart labels.
- **User flow:** **MAKE** hide hazards on the plot → **TEACH** label chart shapes → **TUNE** balance speed/sensitivity against the clock → **COME ALIVE** chest sensor pulses, champion scans → **PROVE** a fresh randomized plot under time → **SHOW OFF** survey the City building site (safe zones glow green) → **KEEP** chest sensor.
- **Done when:** clear a *fresh* grid within the timer at ≥90% accuracy.
- **Buddy moments:** *"At speed 8 you finish but accuracy's 84% — deep pipes blur. Drop to 6, or raise sensitivity +2?" · "Your 'cavity' and 'rock' labels are mixed — the broad flat charts are cavities. Re-label these 3?"*
- **City district:** building-site survey.
- **Watch-outs:** all on-screen → also plays on `file://`.

L11 · Watch-Sense — *flag real hazards without crying wolf*

- **Concept:** a detection threshold trades misses vs. false alarms (precision/recall).
- **Endowment:** the champion can watch a live scene and alert on genuine hazards without false alarms.

- **Engine:** top-down intersection sim; CV bounding boxes; tunable-threshold detector; a dashboard counting *Missed Hazards* + *False Alarms* over 30-second rounds.
- **Real-life job:** the child **stages the intersection** — places pedestrians, a sneaky jaywalker, and distractor pigeons — the scenario the champion must judge.
- **Tuning surface:** the alert threshold (optionally per-class) to drive both counters to zero.
- **User flow:** **MAKE** stage the crossing → **TUNE** run 30s rounds, watch both counters → **COME ALIVE** the eye upgrades, boxes appear, champion calls danger → **PROVE** a fresh randomized 30s → **SHOW OFF** the City intersection turns green when safe → **KEEP**.
- **Done when:** a fresh round with **0 misses AND 0 false alarms**.
- **Buddy moments:** "0.3 → 5 pigeon alarms; 0.8 → you missed the jaywalker. Try ~0.55?" · "Safe on cars but missing cyclists — want a separate lower threshold just for bikes?"
- **City district:** intersection / CCTV safety.
- **Watch-outs:** flash-safe boxes; on-screen only.

L14 · Sky-Sense — *spot genuine collision risks among many objects*

- **Concept:** clustering groups nearby objects; sensitivity sets how tight a cluster is.
- **Endowment:** the champion can track many moving objects and flag only true close-calls.
- **Engine:** radar sim (50 seeded moving dots); distance-based clustering; sensitivity slider; cluster highlight.
- **Real-life job:** the child **sets up the airspace** — launches a fleet with a few deliberately converging paths — then tunes what counts as "too close."
- **Tuning surface:** cluster sensitivity/radius — catch real close-calls, don't bubble every plane.
- **User flow:** **MAKE** place 50 planes + a couple of near-misses → **TUNE** adjust sensitivity, watch clusters form → **COME ALIVE** antenna extends, champion highlights danger clusters → **PROVE** a fresh sky → **SHOW OFF** City air-traffic / crowd safety → **KEEP**.
- **Done when:** on a fresh scenario, flag the genuine close-calls with **no noise clusters**.
- **Buddy moments:** "Too tight: 12 red bubbles, mostly noise. Group only planes within 2 rings?" · "You missed the two crossing at the top — nudge sensitivity up one notch?"
- **City district:** air-traffic / crowd safety.
- **Watch-outs:** seeded; flash-safe.

L16 · Mood-Sense — *gauge the city's mood and route issues*

- **Concept:** sentiment scoring reads feeling from words; routing sends issues to the right place.
- **Endowment:** the champion can read feedback, gauge mood, and route issues to the right department.
- **Engine:** offline lexicon/rule sentiment scorer; live word-cloud; a seeded message stream; department bins.
- **Real-life job (most personal):** the child **gathers real feedback** — their own + classmates' honest opinions about their school/city ("the playground swings are broken," "love the new library").
- **Tuning surface:** the sentiment lexicon (+/- words, weights, slang/negation rules) and routing rules (keyword → department).
- **User flow:** **HUNT** gather 8+ real feedback lines → **TEACH** score them; fix misreads ("sick playground" = good) → **COME ALIVE** audio ring glows, word cloud blooms, champion "feels" the mood → **PROVE** a fresh feedback stream, incl. the slang it was taught → **SHOW OFF** City sentiment board lights by department → **KEEP**.
- **Done when:** correctly score + route a fresh batch, including 1–2 tricky slang/negation cases the child taught.

- **Buddy moments:** "'not bad' scored negative because of 'bad' — add a rule that 'not' flips the next word?" · "'sick' next to 'playground' is a compliment — teach it that?"
- **City district:** sentiment board.
- **Watch-outs:** anonymize (first names only); offline lexicon; moderate gathered text.

Cluster B — BODY

L3 · Rover-Legs — *cover an area completely*

- **Concept:** complete-coverage pathfinding — visit every cell, minimize repeats.
- **Endowment:** the champion can move through an area covering every spot, no bumps, no wasteful repeats.
- **Engine:** grid sim with obstacles + dock; movement/collision engine; coverage % + repeat count; a working baseline sweep.
- **Real-life job:** the child **designs their own messy room** — places walls, furniture, a charging dock — then authors the cleaning strategy for it.
- **Tuning surface:** ordered movement rules (*prefer-unvisited* > *straight* > *wall-follow* > *dock-at-low-battery*) + parameters.
- **User flow:** **MAKE** draw the room + obstacles → **TEACH** pick/order rules, run, watch coverage + repeats → **COME ALIVE** legs attach, champion trundles and cleans → **PROVE** a fresh room → **SHOW OFF** patrol/clean a City street → **KEEP**.
- **Done when:** 100% coverage on a fresh layout under a repeat budget, zero collisions.
- **Buddy moments:** "*It re-cleans the middle and skips corners — lift 'prefer unvisited' above 'go straight'?*" · "*Stuck in the U-shaped nook — add 'wall-follow when boxed in'?*"
- **City district:** street cleaning / patrol.

L4 · Reflex-Wiring — *react to sensor events with if-then reflexes*

- **Concept:** a behavior tree / conditional logic with loops and branches.
- **Endowment:** the champion can react to live sensor events with if-then reflexes.
- **Engine:** dense-forest sim with moving obstacles; flight engine; a palette of condition + action nodes; a live flowchart tracer that lights up and flags where a loop breaks.
- **Real-life job:** the child **designs the obstacle course** (trees, birds, wind gusts) then **wires** the champion's reflexes to survive it.
- **Tuning surface:** wire condition→action nodes; reorder; repair broken loops; handle combos (bird *and* tree; low battery).
- **User flow:** **MAKE** build the course → **WIRE** connect `[If Tree]→[Fly Up]→[Resume]`, run, watch the flowchart light up → **COME ALIVE** nervous-system wiring glows, champion reacts live → **PROVE** a fresh course with new combos → **SHOW OFF** autonomous obstacle handling → **KEEP**.
- **Done when:** complete a fresh course without crashing; the loop escapes correctly.
- **Buddy moments:** "'Resume' never fires when a bird's also above — add an `[If Bird]→[Wait]` before it?" · "*Your loop never ends if the tree stays — add a timeout branch?*"
- **City district:** autonomous obstacle handling.

L5 · Twin-Drones — *command helpers and split the work*

- **Concept:** multi-agent task allocation under constraints.
- **Endowment:** the champion can command helper units and divide work among them.
- **Engine:** farm sim (seeded ripen/spoil events); two drones with distinct speed/capacity; a policy-execution engine; live score.
- **Real-life job:** the child **designs the orchard** (tree positions, ripen timings) then writes the division-of-labor policy.
- **Tuning surface:** allocation rules (fast → nearest-single, big → clusters); return-to-base fill thresholds; zones.
- **User flow:** **MAKE** lay out the orchard + timings → **TEACH** set the two-drone policy, run, watch harvest vs. spoilage → **COME ALIVE** shoulder bay opens, two drones deploy → **PROVE** a fresh orchard → **SHOW OFF** mini delivery fleet → **KEEP**.
- **Done when:** beat a harvest target on a fresh orchard with minimal spoilage.
- **Buddy moments:** *"Fast drone chases far clusters and fruit spoils — give clusters to the big one?"* · *"Both idle at base at once — stagger their return thresholds?"*
- **City district:** mini delivery fleet.

L6 · Haul-Planner — *plan an efficient loop under limits*

- **Concept:** constrained route optimization (capacity + weight penalty); compare to an AI baseline.
- **Endowment:** the champion can plan an efficient collection loop under load & battery limits.
- **Engine:** town map with seeded stops; truck physics (weight → speed/battery); route executor; an AI-optimal baseline to reveal.
- **Real-life job:** the child **places the collection stops** (their own town) then plans the loop + reload strategy.
- **Tuning surface:** the ordered route + when to return to depot; distance vs. weight penalty.
- **User flow:** **MAKE** drop stops → **TEACH** draw the loop + reloads, run, watch fuel/battery/time → **PROVE-vs-AI** hit "Run AI" to overlay the optimal → **COME ALIVE** back module installs, champion runs it → **SHOW OFF** City waste/delivery route → **KEEP**.
- **Done when:** complete all stops within battery on a fresh map, within X% of the AI-optimal.
- **Buddy moments:** *"20 bags uphill drains battery — reload after stop 3?"* · *"You beat the AI on distance but lost on weight — drop the heavy stop earlier."*
- **City district:** waste / delivery routes.

Cluster C — PLANNER

L7 · Transit-Brain — *schedule a network to cut waiting*

- **Concept:** network transfer timing across intersecting routes.
- **Endowment:** the champion can schedule a transport network to minimize waiting.
- **Engine:** 3-intersecting-route sim; passengers; transfer/wait clock; timetable executor.
- **Real-life job:** the child **surveys their real class** — "what time do you arrive / go home?" — building the actual demand curve the system must serve.
- **Tuning surface:** departure times + frequency per route to minimize transfer waiting for *that* demand.

- **User flow:** **HUNT** run a quick class survey → **TEACH** set timetables to the demand, run, watch total wait → **COME ALIVE** brain core boots, champion runs the timetable → **PROVE** a fresh demand day → **SHOW OFF** City bus timetable → **KEEP**.
- **Done when:** average wait under a target on a fresh demand pattern.
- **Buddy moments:** *"Class lands 8:00–8:15 but Blue runs empty at 7:30 — shift it later?" · "6-min wait at the transfer — sync Green to arrive 2 min after Red?"*
- **City district:** bus timetable.
- **Watch-outs:** survey = times only, anonymized.

L8 · Grow-Planner — *place infrastructure before it's needed*

- **Concept:** forecast-driven placement; act on predictions, not the present.
- **Endowment:** the champion can forecast demand and place infrastructure ahead of need.
- **Engine:** 10-year fast-forward timeline; population/aging heatmap projector; building types (slow Hospital vs. instant Tech-Clinic); budget.
- **Real-life job:** the child **designs their own neighborhood** (homes, parks, people) — their city to care for.
- **Tuning surface:** what to build where **and when**, reading the forecast, under budget.
- **User flow:** **MAKE** lay out the neighborhood → **TEACH** read the forecast, place builds on a timeline under budget → **COME ALIVE** planner core lights, the decade fast-forwards → **PROVE** did zones get care before they aged/collapsed? → **SHOW OFF** City long-term planning → **KEEP**.
- **Done when:** all zones stay served across a fresh 10-yr run without bankruptcy.
- **Buddy moments:** *"North ages in ~5 yrs and a hospital takes 5 to build — start it now?" · "You spent it all on hospitals — a cheap Tech-Clinic covers that small spike better."*
- **City district:** long-term city planning.

L9 · Flow-Planner — *shift load and hold a reserve*

- **Concept:** load-shifting schedule; move demand to off-peak, keep a safety reserve.
- **Endowment:** the champion can schedule resource pumping to shift load and hold a reserve.
- **Engine:** pumps + reservoirs network; off-peak/peak energy pricing; reservoir levels; an emergency-reserve line.
- **Real-life job:** the child **maps their city's daily water rhythm** (or surveys real home habits: when does your family shower/cook?).
- **Tuning surface:** the pump on/off schedule → cheap off-peak while staying above the reserve line.
- **User flow:** **HUNT** map the daily usage pattern → **TEACH** build the pump schedule, run a day, watch cost + level → **COME ALIVE** planner core, champion runs the pumps → **PROVE** a fresh day with a surprise spike → **SHOW OFF** City water/energy timing → **KEEP**.
- **Done when:** finish a fresh day under a cost target, never dipping below the emergency reserve.
- **Buddy moments:** *"Pumping at 6pm peak is priciest — move to 2am?" · "Careful — off-peak-only leaves the reservoir dry when the morning spike hits."*
- **City district:** water & energy timing.

L10 · Grid-Warden — *govern autonomous agents with guardrails*

- **Concept:** setting guardrails/rules over autonomous AI bidding agents — oversight of AI.
- **Endowment:** the champion can govern autonomous AI agents with fairness guardrails.

- **Engine:** a market of autonomous bidding agents (data centers vs. homes vs. schools); dynamic pricing; supply events.
- **Real-life job:** the child **writes the fairness rules** — their policy for who gets power when supply is short.
- **Tuning surface:** pricing + guardrails (a floor guaranteeing schools/hospitals power no matter the corporate bid).
- **User flow:** **DECIDE** set guardrails + pricing → **TEACH** run the market, watch who wins when supply drops → **COME ALIVE** warden core, champion referees → **PROVE** a fresh shortage event → **SHOW OFF** City power market → **KEEP**.
- **Done when:** survive a fresh supply crunch with schools/hospitals always above their floor.
- **Buddy moments:** *"Clouds hit solar → data centers outbid the school and it goes dark. Set a 20% floor for schools?"* · *"Your floor's so high the market can't clear — try 15%."*
- **City district:** power market.
- **Watch-outs:** "AI agents" explained honestly as automated bidders.

L13 · Ripple-Warden — *balance a system watching side effects*

- **Concept:** second-order effects — fixing one thing spills into another.
- **Endowment:** the champion can balance a system while watching knock-on side effects.
- **Engine:** traffic-wave sim; reroute tools (green-flush / detour); *Echo Gauges* on residential streets showing spillover.
- **Real-life job:** the child **designs their street grid** and picks which avenue to rescue.
- **Tuning surface:** green-light ratios across blocks so a fix doesn't flood the next street.
- **User flow:** **MAKE** draw the grid + a jam → **TEACH** reroute, watch the Echo Gauges → **COME ALIVE** warden core, champion balances flow → **PROVE** a fresh jam → **SHOW OFF** City-wide traffic flow → **KEEP**.
- **Done when:** clear a fresh jam keeping all Echo Gauges under the red line.
- **Buddy moments:** *"You cleared Main but Oak's Echo Gauge hit 80% — bleed 30% back to 5th?"* · *"Nice — spreading it over 3 streets kept every gauge green."*
- **City district:** city-wide traffic flow.

Cluster D — CONSCIENCE (the capstone arc)

L12 · Judgment-Core — *decide by learned values*

- **Concept:** preference learning — the AI imitates the human's judgment.
- **Endowment:** the champion can make value-based decisions it learned from *your* judgment.
- **Engine:** a dilemma-scenario generator; a "learn from your choices" model; a test mode on fresh random conditions.
- **Real-life job:** the child's **own judgment calls** are the training data (rain + school-out → protect the buses).
- **Tuning surface:** make the calls (build a values dataset); watch the AI imitate; correct where it mislearned.
- **User flow:** **MAKE** answer dilemma scenarios → **TEACH** the Core learns, then runs on fresh conditions → **COME ALIVE** heart installs, champion decides "like you" → **PROVE** does it choose as you would on unseen dilemmas? correct it → **SHOW OFF** City ethical priority calls → **KEEP**.

- **Done when:** the Core matches the child's held-out choices above a threshold, and the child reviews/fixes a surfaced bias.
- **Buddy moments:** *"You always protect kids over speed — but in fog you once picked speed. On purpose, or should I learn 'safety in fog'?" · "It learned to always favor buses — even at 3am with none. Add a 'time of day' example?"*
- **City district:** ethical priority calls.
- **Watch-outs:** surfaces bias honestly; the child stays the author of the values.

L15 · Cost-Sense — ***pick the right AI tool within a budget (generative-AI literacy centerpiece)***

- **Concept:** model selection (speed/cost/accuracy); prompting; compute cost.
- **Endowment:** the champion can pick the right AI tool for a job within a compute budget.
- **Engine:** three AI engines differing in speed/cost/accuracy; a prompt→generate demo (bronze cheap/blocky vs. gold rich); a budget meter.
- **Real-life job:** the child **writes their own prompts** and chooses which engine to buy for a real task (design a park / answer citizen questions).
- **Tuning surface:** match engine to task; spend compute wisely; write prompts that get good results cheaply.
- **User flow:** **MAKE** get a task + budget, write a prompt → **TEACH** try engines, see cost vs. quality → **COME ALIVE** Cost-Sense heart, champion chooses tools wisely → **PROVE** a fresh task under a tighter budget → **SHOW OFF** City cost-aware AI services → **KEEP**.
- **Done when:** complete a fresh task above a quality bar under budget, choosing engines well.
- **Buddy moments:** *"Vague prompt → the cheap engine gave mush. Add 3 detail words and retry cheap before a gold coin?" · "You used gold for a plain text reply — bronze nails that; save gold for images."*
- **City district:** cost-aware AI services.
- **Watch-outs:** bounded, teacher-keyed, content-filtered generation.

L17 · Auditor-Core — ***oversee an imperfect AI (human-in-the-loop; the final endowment)***

- **Concept:** human-in-the-loop; anomaly detection; the cost of false positives.
- **Endowment:** the champion can oversee an imperfect AI and catch its mistakes.
- **Engine:** a conveyor of profiles; an imperfect AI flagger; an "AI says X vs. reality Y" cross-reference; false-positive fallout (an angry wrongly-frozen citizen); an approval-rating meter.
- **Real-life job:** the child **is the human-in-the-loop** — verify the AI's flags, stamp release/freeze, own the fallout.
- **Tuning surface:** how much to trust the AI; handle false positives before approval tanks.
- **User flow:** **DO** cross-reference flagged profiles, stamp release/freeze → **TEACH** corrections tune trust; false freezes anger citizens → **COME ALIVE** Auditor heart, champion audits with your judgment → **PROVE** a fresh stream: catch real fraud, spare the innocent, keep approval up → **SHOW OFF** City finance auditor → **KEEP** — *champion complete*.
- **Done when:** a fresh audit round with real fraud caught, false positives handled, approval above target.
- **Buddy moments:** *"AI flagged the bakery but the numbers match — freezing it angers a real citizen. Double-check before you stamp?" · "You're rubber-stamping the AI — it's wrong ~1 in 5. Slow down on the big transfers."*
- **City district:** finance auditor.

- **Watch-outs:** teaches AI fallibility honestly; caps the conscience arc.
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Capstone · Champion Reveal Day (design + present)

- **Real-life job:** finalize the humanoid's design and prepare a presentation.
 - **Flow:** **FINALIZE** look, name, story (Champion Creator) → **WALK** the champion tours the AI City, demoing all 17 powers district by district → **PRESENT** the child shows it off: name, design choices, what each part does, one proud moment → **BUDDY** helped rehearse talking points beforehand, then stepped back → **KEEP** export the Champion Card + an optional recorded/printed reveal.
 - **Why it lands:** synthesis + public-speaking + pride; and a genuine gen-AI-literacy capstone (using the buddy as a tool to prepare *your own* work, not to do it).
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Open content questions for review

1. **L2 / L11 / L14 / L13** — is "the child authors the test scenario" the right own-content hook, or would you rather they tune against pre-set scenarios? (I chose authoring for ownership.)
2. **L7 / L9 surveys** — comfortable with a light real-class/home survey (times/habits only, anonymized), or keep it fully in-sim?
3. **Done bars** — the concrete thresholds ($\geq 90\%$, "within X% of AI-optimal," approval targets) are placeholders to be pinned per project at build time.
4. **Project length** — each is scoped as a ~single-session follow-on to its lesson; confirm that fits the timetable.